

Solar Thermochemical Heat Export via Closed-Loop Salt Hydration

Eric Neuman

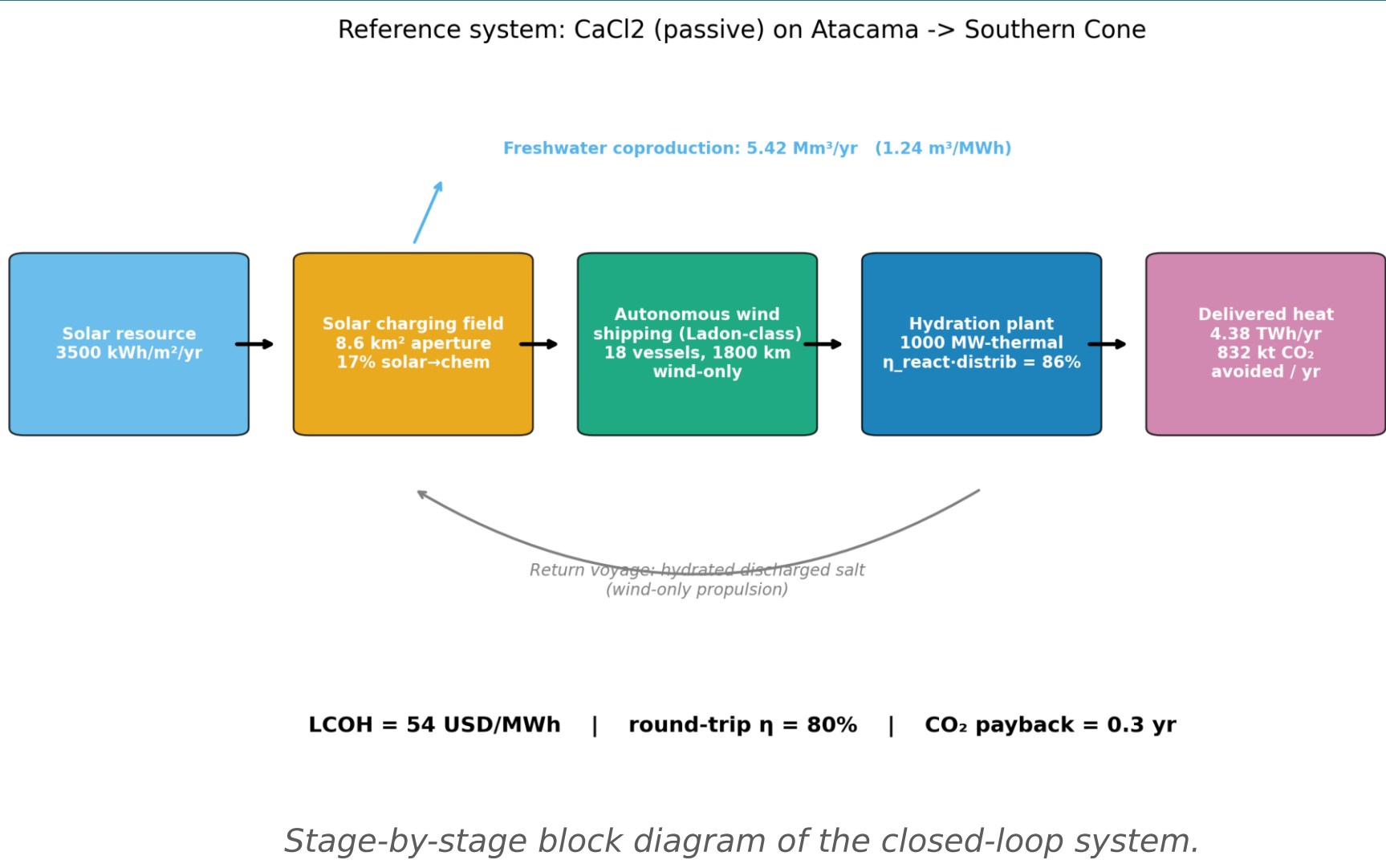
Charging desert salt with sunlight, shipping it on autonomous wind-primary freighters, releasing heat in dirty-grid winter cities.

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TL;DR

- Charge inorganic salt hydrates with desert sun. Ship the dry salt to cold cities on uncrewed wing-sail freighters. Release stored heat by hydrating the salt at a district-heating plant. Return the wet salt for recharge.
- Two viability tiers in v0.3:
 - (a) short-haul + passive pond + $\text{CaCl}_2 \cdot 6\text{H}_2\text{O} \leftrightarrow \text{CaCl}_2 \cdot 2\text{H}_2\text{O}$
 - (b) long-haul + linear-Fresnel CSP + $\text{CaCl}_2 \cdot 6\text{H}_2\text{O} \leftrightarrow \text{CaCl}_2$
- Replacing cargo-as-fuel propulsion with autonomous wind-primary shipping is the v0.3 architectural pivot. It cuts long-haul LCOH nearly in half and unlocks the highest-CO₂-leverage routes.

REFERENCE SYSTEM — ATACAMA → SOUTHERN CONE



HEADLINE NUMBERS (END-TO-END SIMULATION)

	Short-haul best	Long-haul best
Working fluid	CaCl_2 passive	CaCl_2 deep CSP
Route	Atacama → S. Cone	Pilbara → N. China
Distance	1,800 km	8,000 km
LCOH	53.8/MWh — $\sim$ 100.6 / MWh-th	
Round-trip thermal η	80%	72%
CO ₂ avoided / 1 GW-th yr	832 kt	1,533 kt
Solar field area	8.6 km² (pond)	6.2 km² (CSP)
Freshwater coproduction	5.4 Mm³/yr	2.9 Mm³/yr
Autonomous fleet size	18 vessels	31 vessels
Round-trip cycle days	16 d	56 d

WHAT IS NEW

Closed-loop maritime thermochemical heat export has not, to our knowledge, been previously modeled end-to-end. Adjacent literature treats either (i) stationary salt storage at the plant, or (ii) hydrogen / ammonia carriers.

Salt-hydrate carriers are uniquely suited to low-temperature district heat: the discharge temperature is the use temperature, with no exergy-destroying conversion to and from electricity or chemical fuel.

METHODS

Two-track model:

- Modelica package (SolarSaltExport/) — declarative, acausal, the canonical specification of the energy and mass balances at each stage.
- Python mirror (sim/) — 1:1 correspondence to the Modelica components, used as the numerical runner for parameter sweeps and figure generation. Stages: solar charging field → outbound voyage → hydration reactor → district-heat plant → return voyage.
- Six candidate chemistries × six routes evaluated on a common grid. Single largest residual uncertainty: multi-cycle salt degradation.

CANDIDATE WORKING FLUIDS

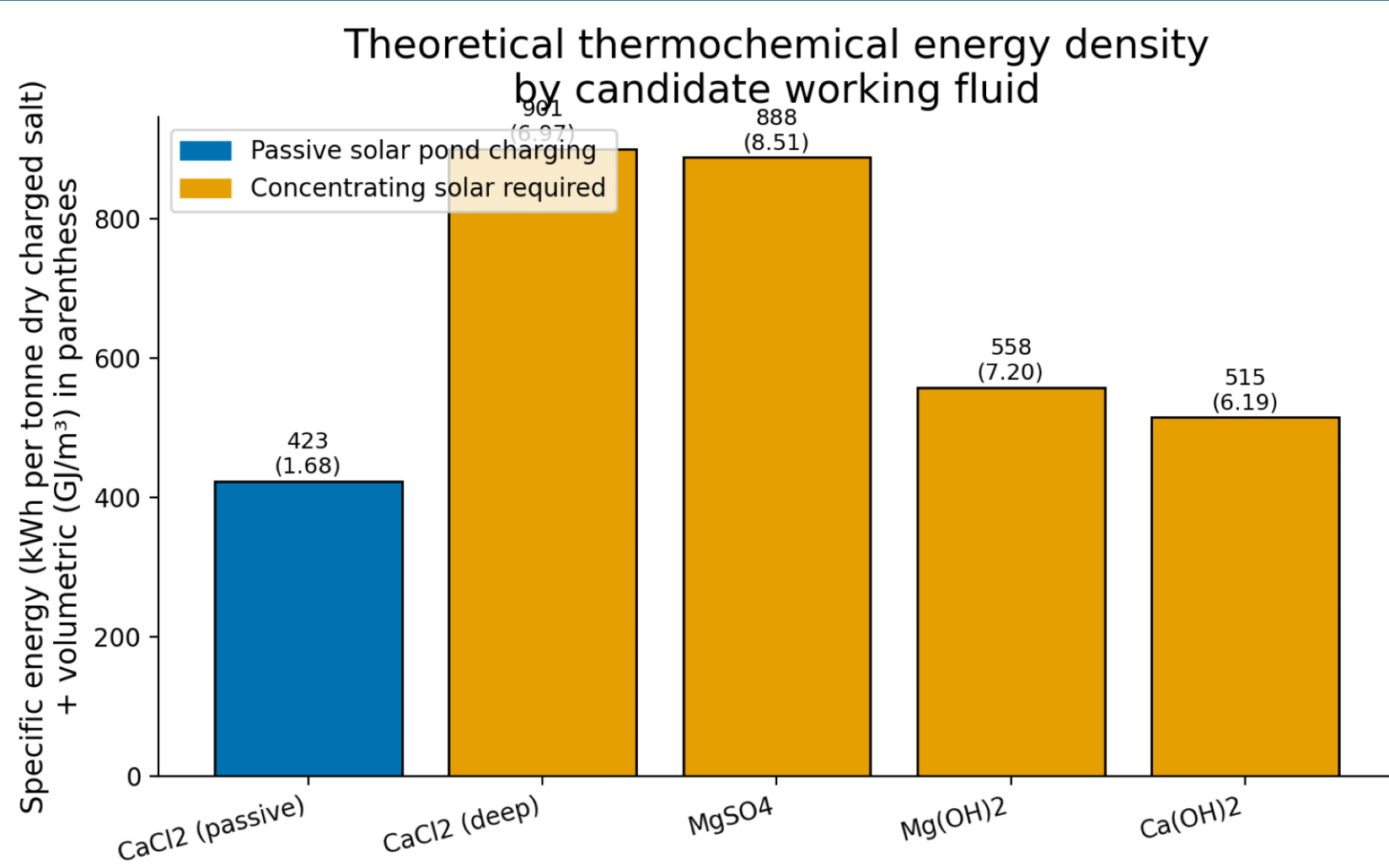


Fig. 1 — Theoretical energy density of candidate salt-hydrate pairs.

CHEMISTRY × ROUTE GRID

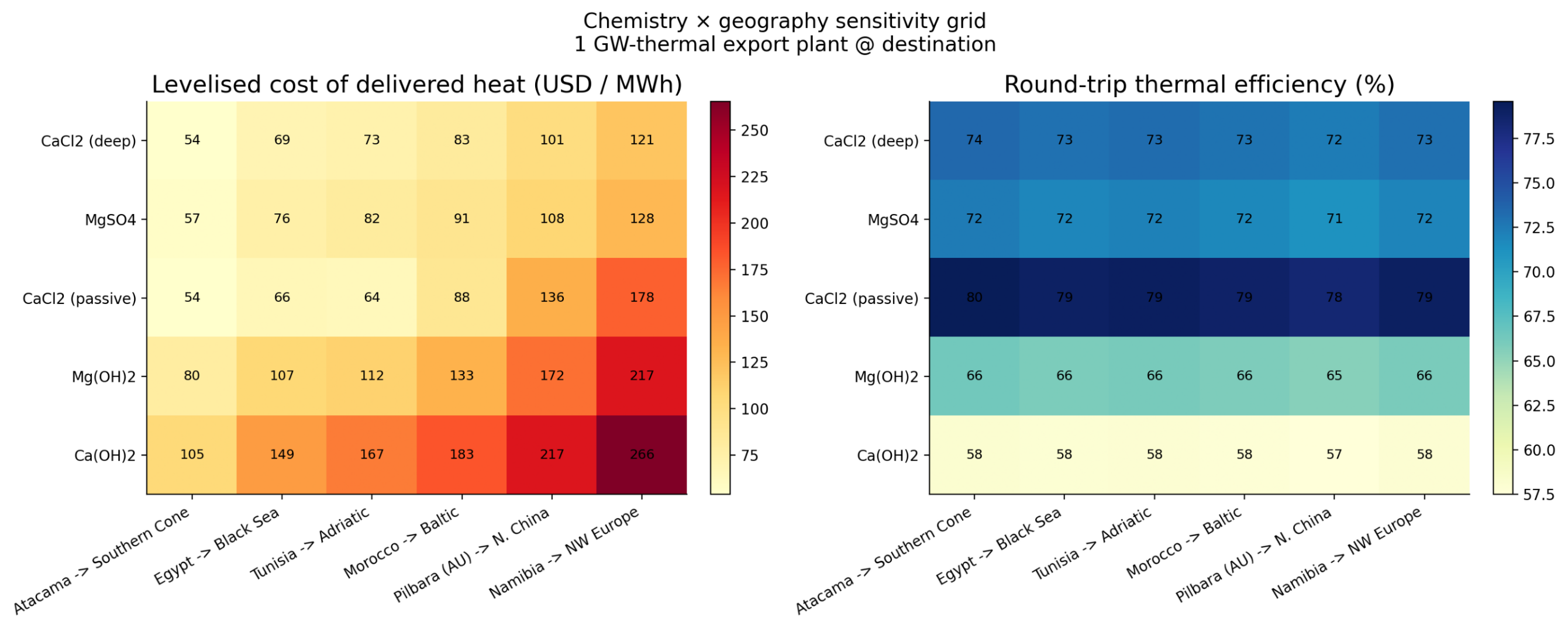


Fig. 2 — LCOH (left) and round-trip thermal efficiency (right) across the chemistry × route matrix.

100 UNITS DNI → DELIVERED HEAT

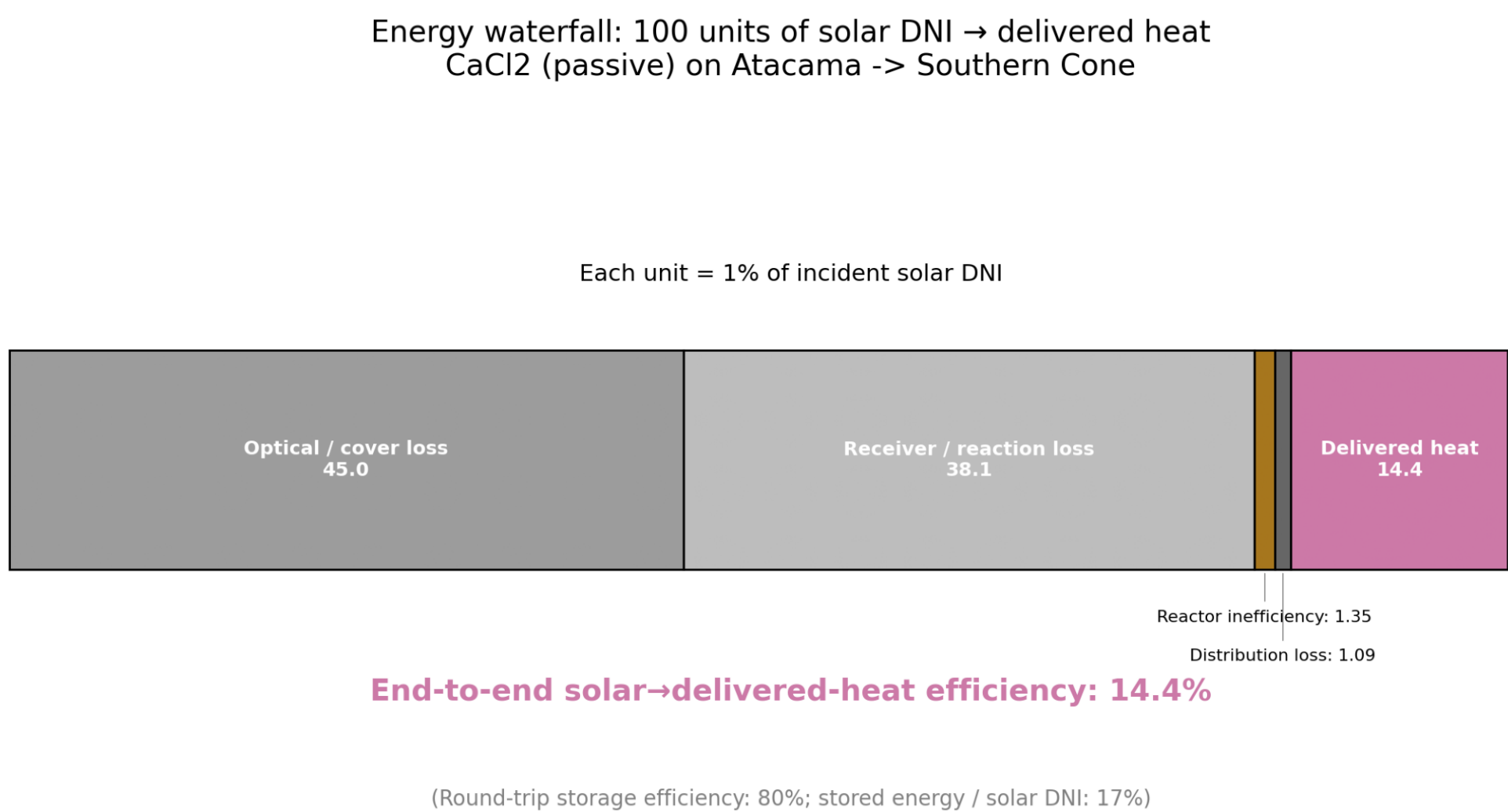


Fig. 10 — Energy waterfall from solar resource to delivered heat.

LCOH VS AUTONOMOUS-VESSEL CRUISE SPEED

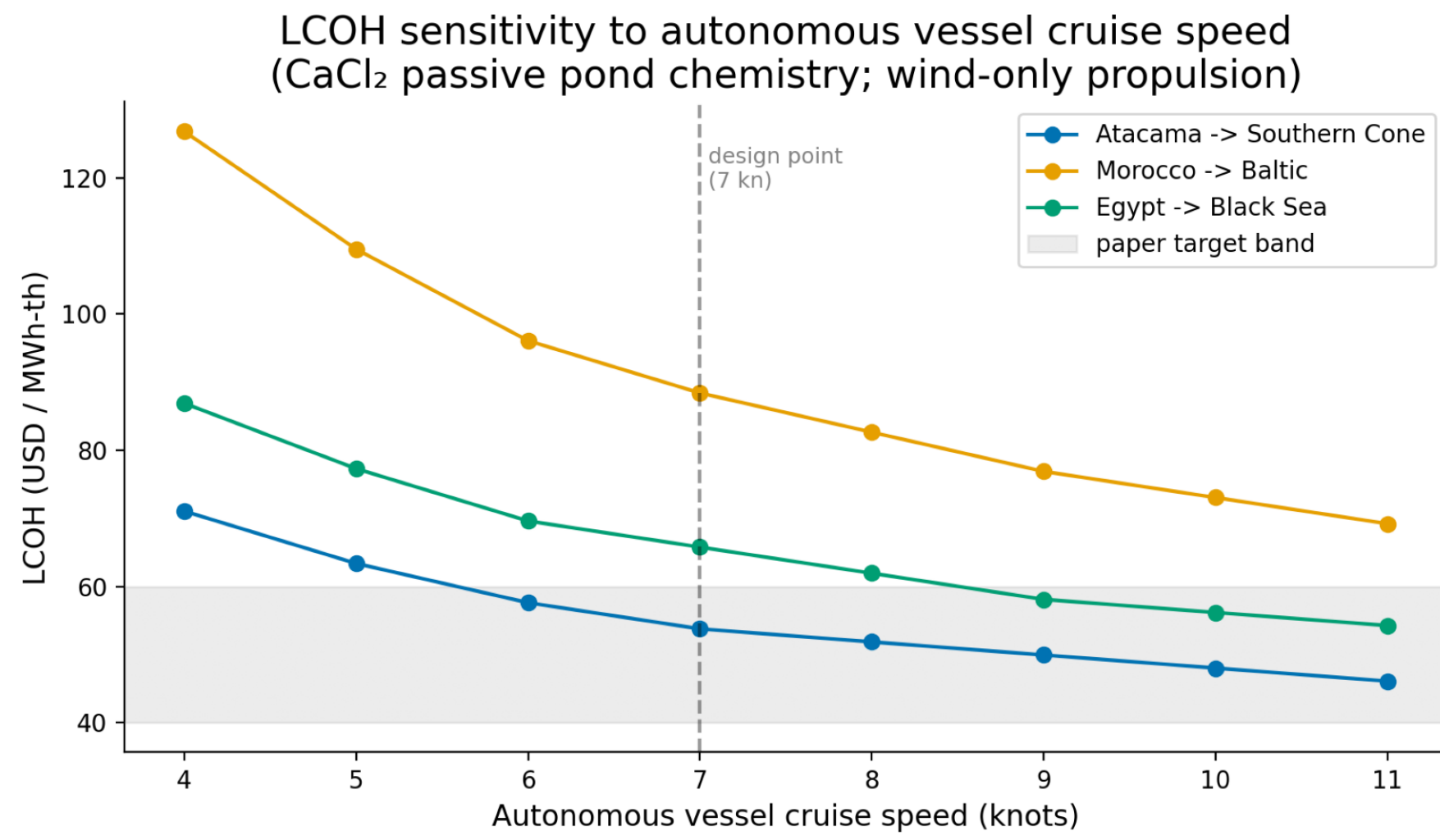


Fig. 4 — Sensitivity of LCOH to cruise speed across the route set ($\approx$ \$5 / MWh per knot).

CLIMATE LEVERAGE

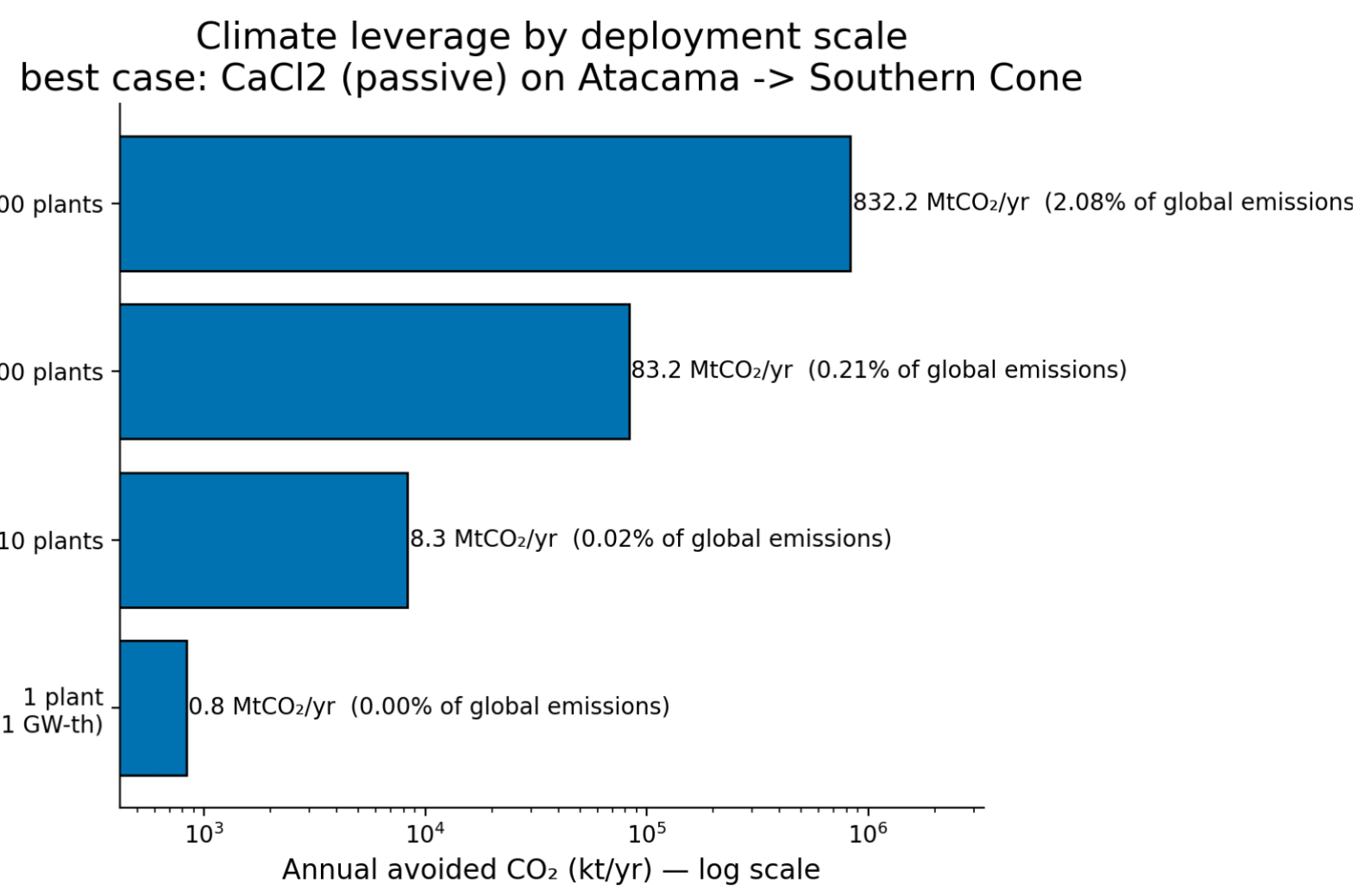


Fig. 8 — Cumulative CO₂ avoided vs deployment scenario.

DEPLOYMENT VS IEA NZE WEDGE

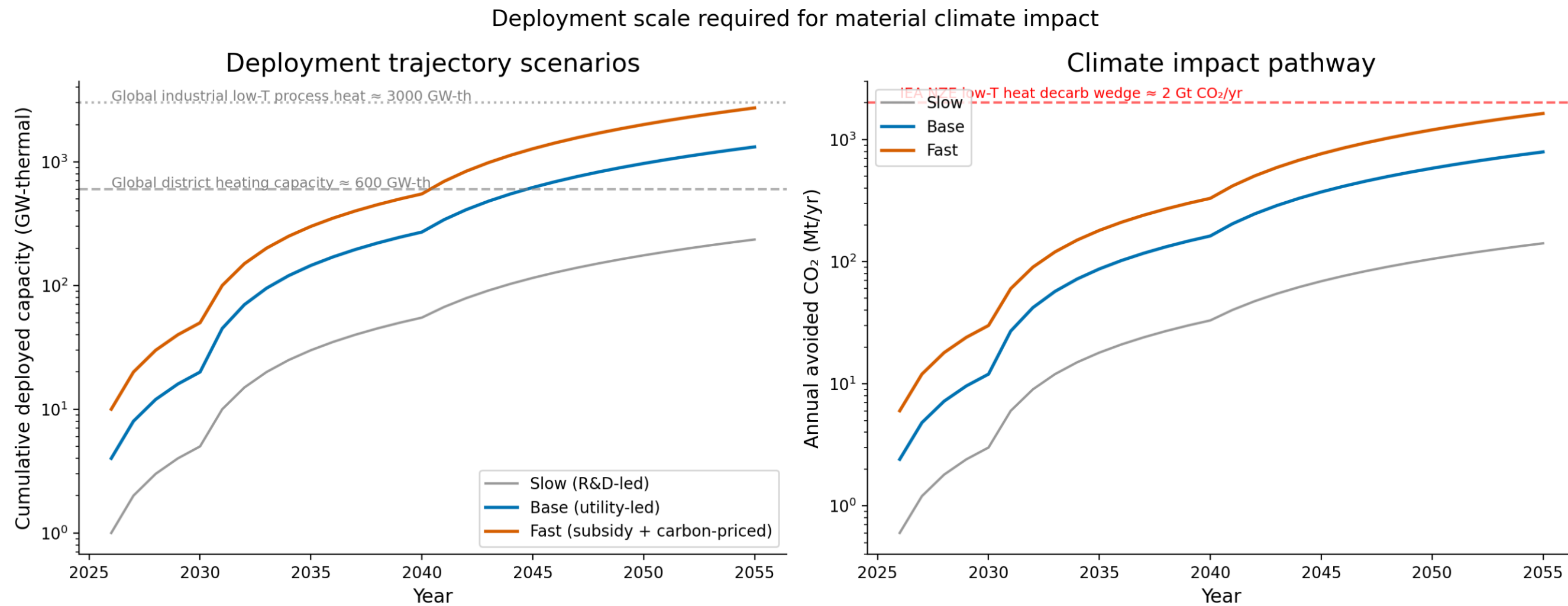


Fig. 11 — Fast scenario reaches $\approx$ 80% of the IEA NZE low-T heat wedge by 2050.

FINDINGS & NEXT STEPS

Findings

- Two viability tiers, not one. Short-haul passive-pond CaCl_2 clears 54–66/MWh; long-haul deep — CSP CaCl_2 sits at 100–120/MWh.
- Cargo-as-fuel was the v0.2 showstopper. Autonomous wind-primary shipping eliminates it: round-trip efficiency rises to a uniform 58–80%.
- Freshwater coproduction (2–6 Mm³/yr per GW-th) is a real, monetisable co-revenue stream in water-scarce source regions.
- Achievable wedge $\approx$ 1.6 Gt CO₂/yr — $\approx$ 80% of the IEA NZE low-T heat decarbonization gap.

Binding open questions

- Multi-cycle salt stability ($\geq$ 1,000 cycles). Bench item.
- Autonomous wind-cargo fleet scale-up (5,000+ ships by 2050).
- Site-specific wind statistics in cruise-speed model.

Next steps

- Engage Ladon Robotics for vendor-grade vessel specs.
- 200-cycle CaCl_2 bench at a partner lab.
- 100 kW-th pilot at Mejillones, Chile with a single autonomous vessel.

Selected references

N'Tsoukpoe et al. 2009; Donkers et al. 2017; Trausel et al. 2014; Cot-Gores et al. 2012; IEA NZE 2023.